

FROM HE TO HOLMES: DECODING THE NEURAL REPRESENTATION OF COREFERENCE



WHAT IS COREFERENCE?

—



To Sherlock Holmes she is always the women.

He felt no emotion akin to love for Irene Adler.

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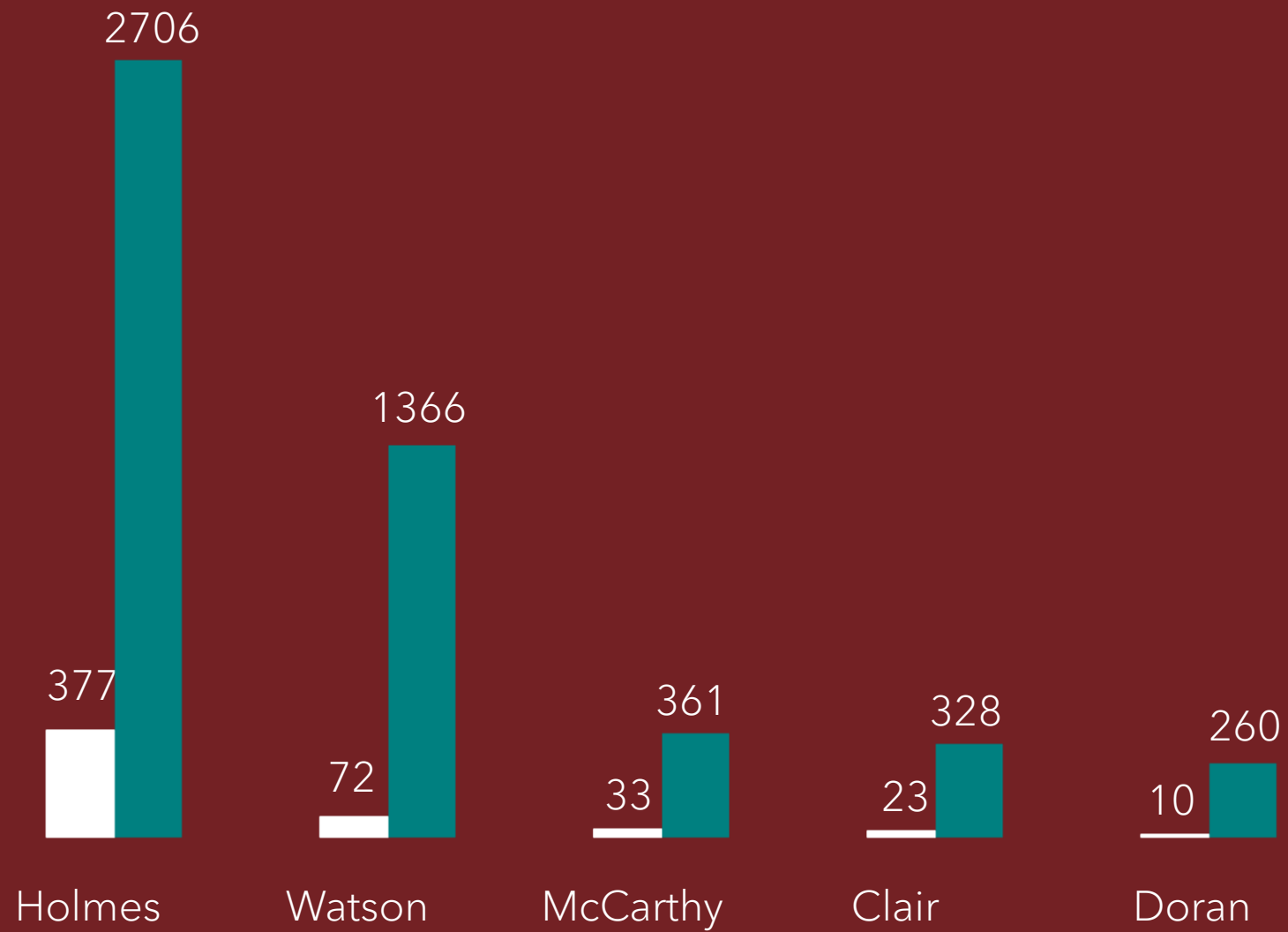
He felt no emotion akin to love for Irene Adler.

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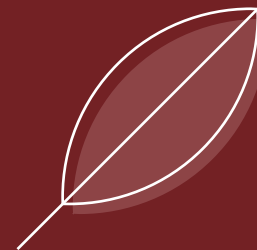


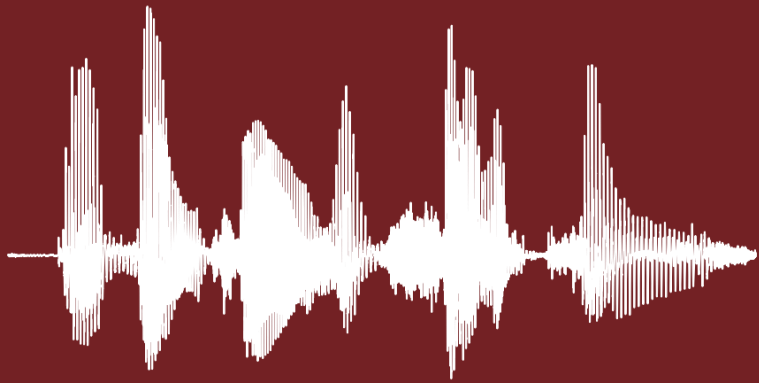
Frequency of Character Names



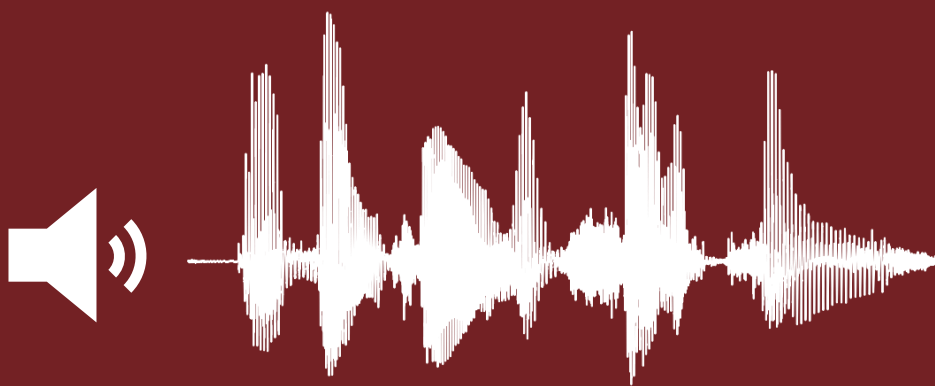
Frequency of Character References

EXPERIMENTAL DESIGN

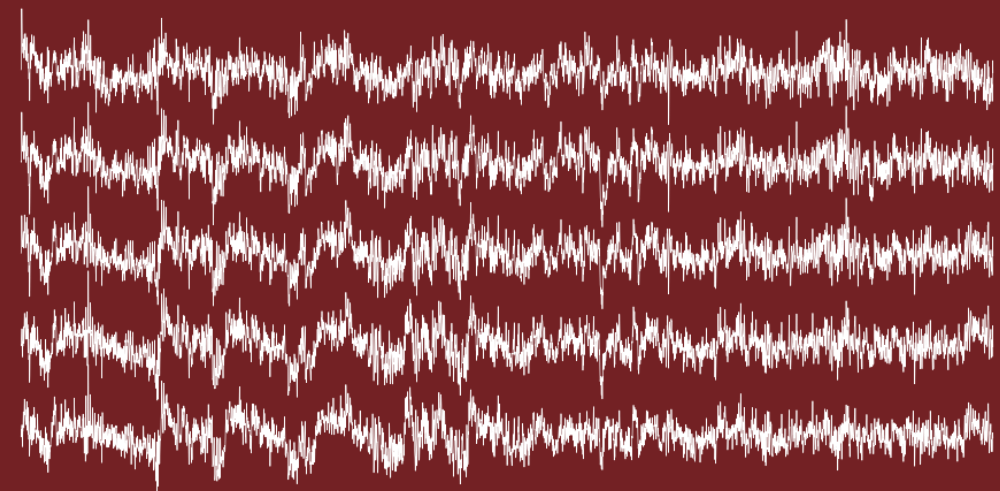


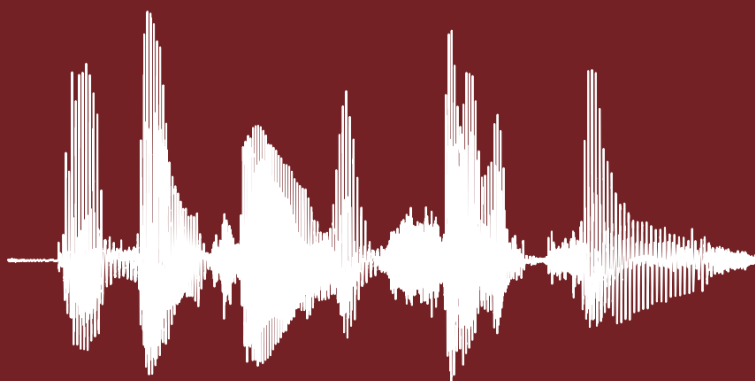


To Sherlock Holmes she is always the women.

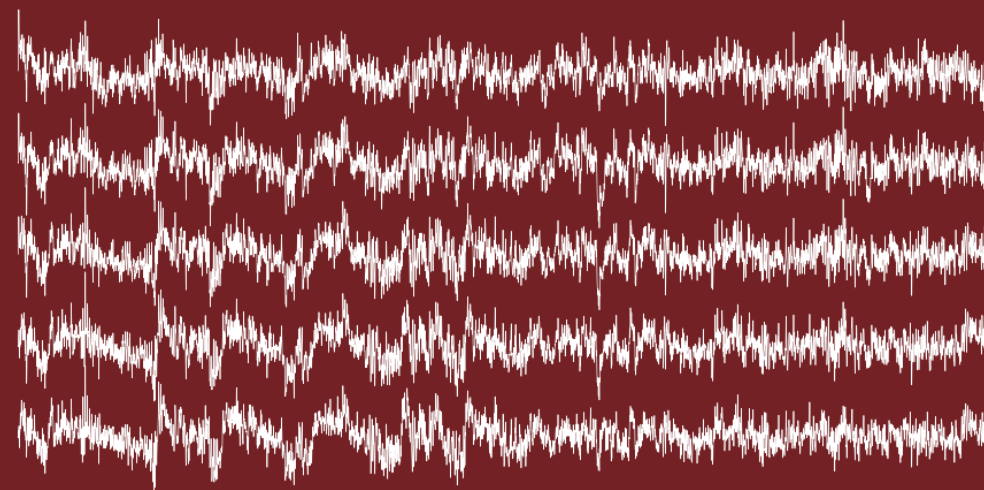


To Sherlock Holmes she is always the women.





3 patients



To Sherlock Holmes she is always the women.

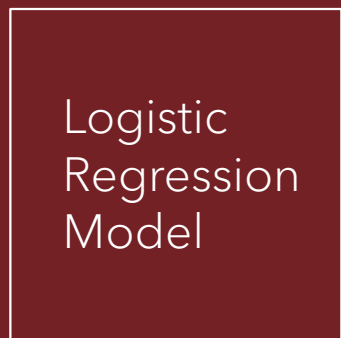
PROPN

PROPN

PRON

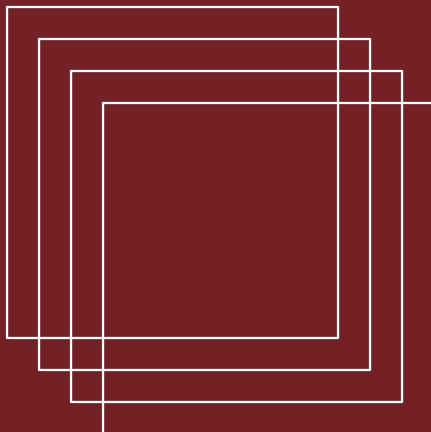
NOUN

Features: MEG Channels

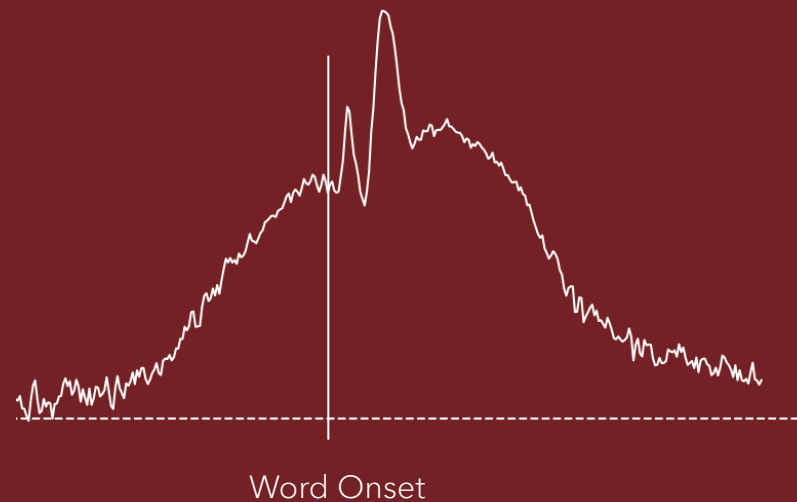


Labels: Semantic Features

401 Time Points



ROC
AUC
Score



ANALYSIS 1:
ARE DIFFERENT TYPES OF
NOUNS DIFFERENTIABLE?



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- 1) Segmented the story into 5 samples
- 2) Used Trained 1 v All Logistic Regression Model with part of speech as the classes
- 3) Used ROC AUC to evaluate decodability
- 4) Used a permutation-cluster test to establish statistical significance

Decoding Performance over Time (AUC)

0.60

0.55

0.50

-2s

0s

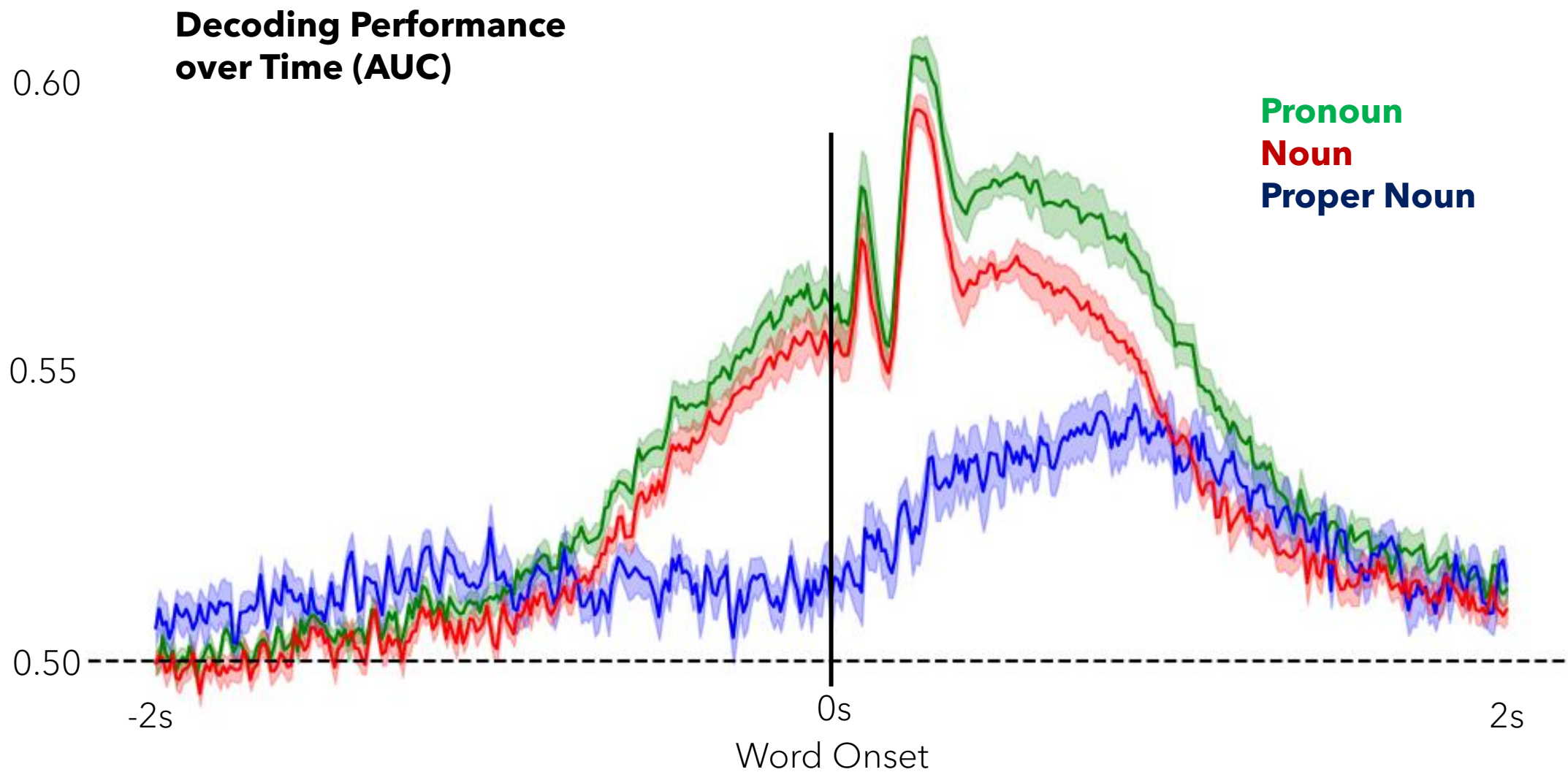
2s

Word Onset

Pronoun

Noun

Proper Noun



ANALYSIS 2: ARE THERE
SEMANTICALLY DIFFERENT
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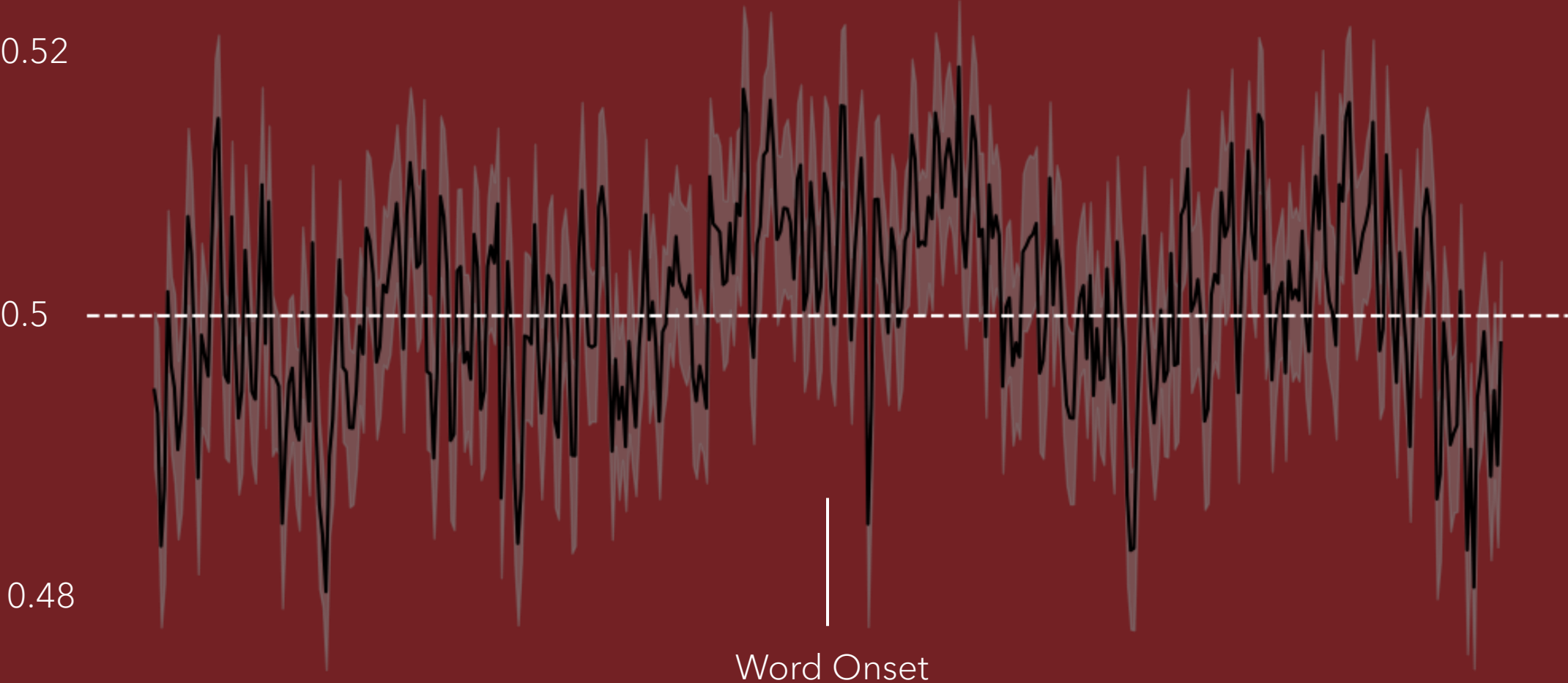
1) Selected the 18 most common characters in the dataset

2) Selected the 10 most common pronouns associated with these characters

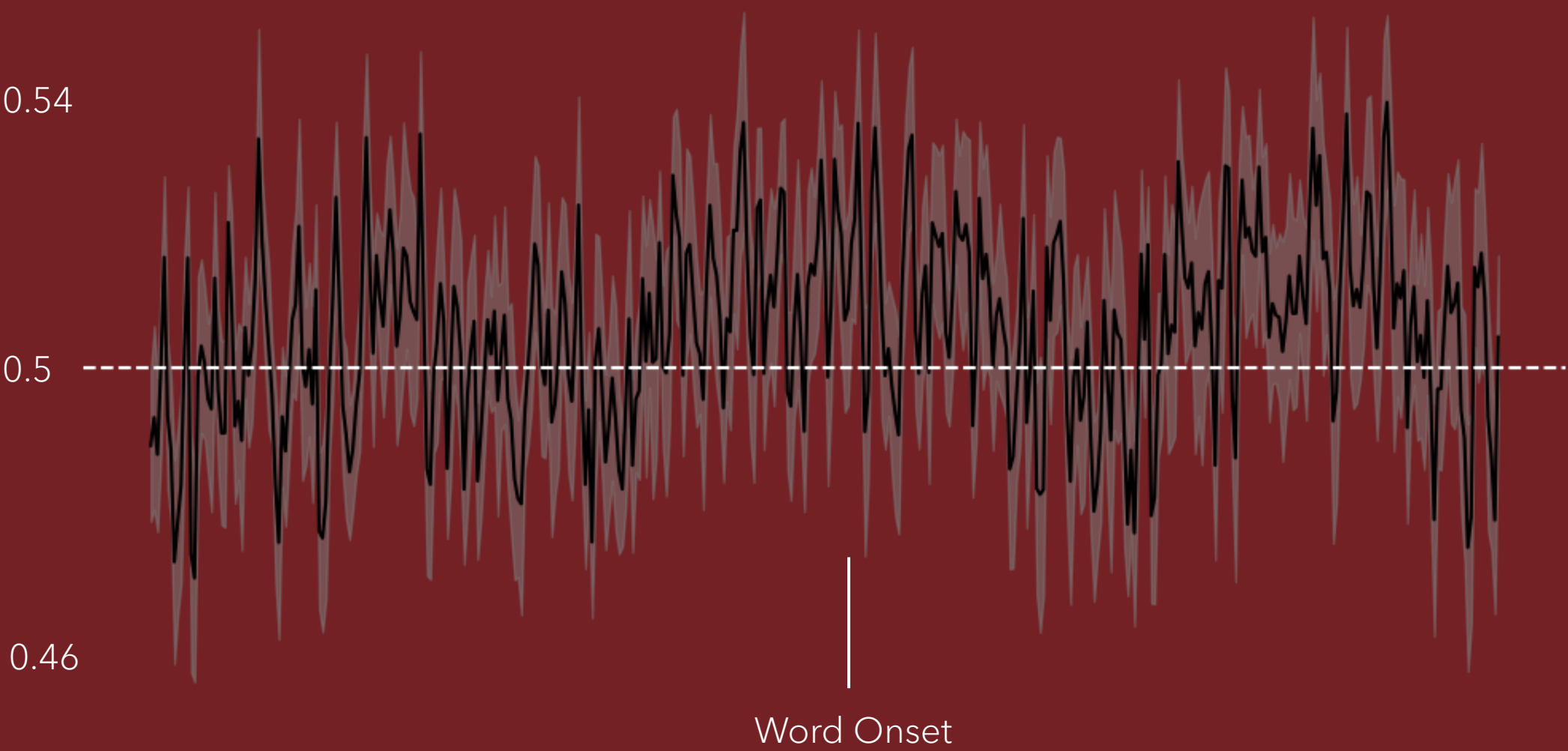
3) For each pronoun, trained a 1 v all LR model for each character with at least 30 coreferences

4) Evaluated using same metrics

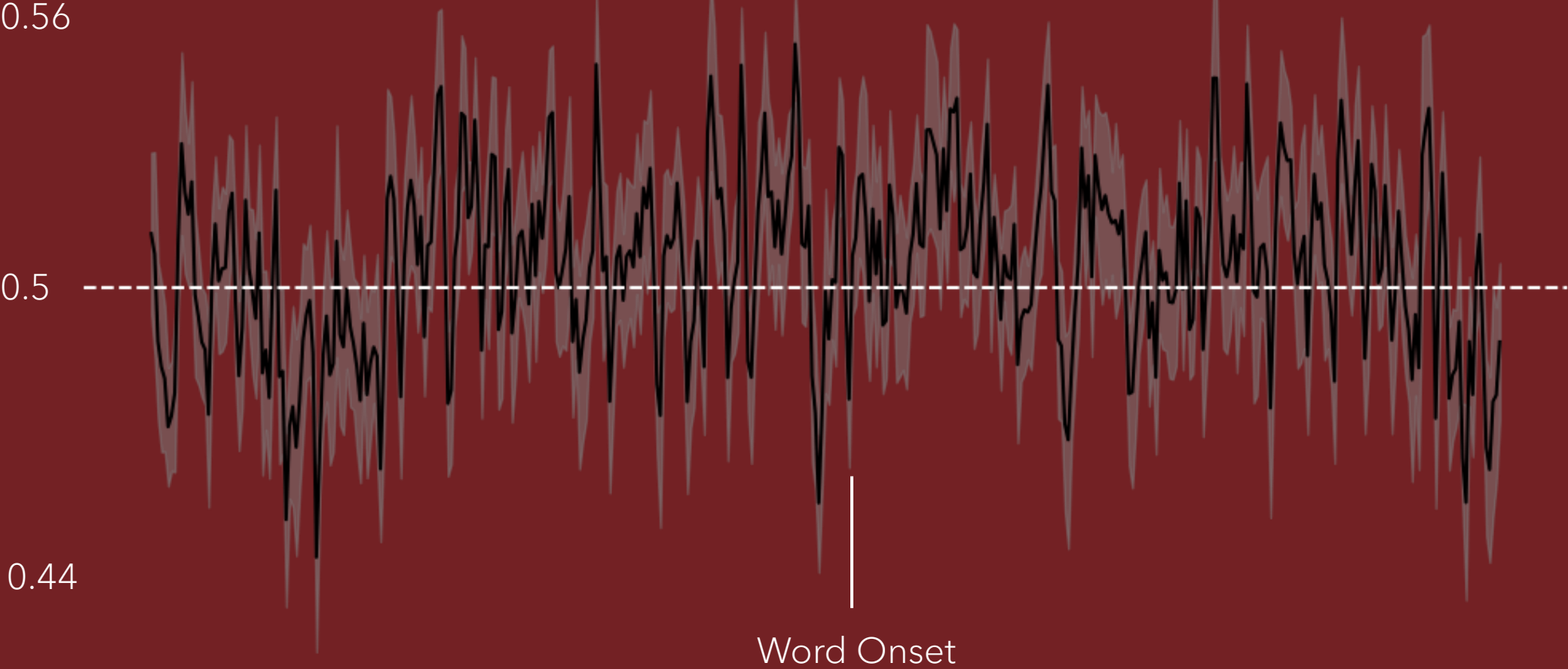
All: Decoding Performance over Time (AUC)



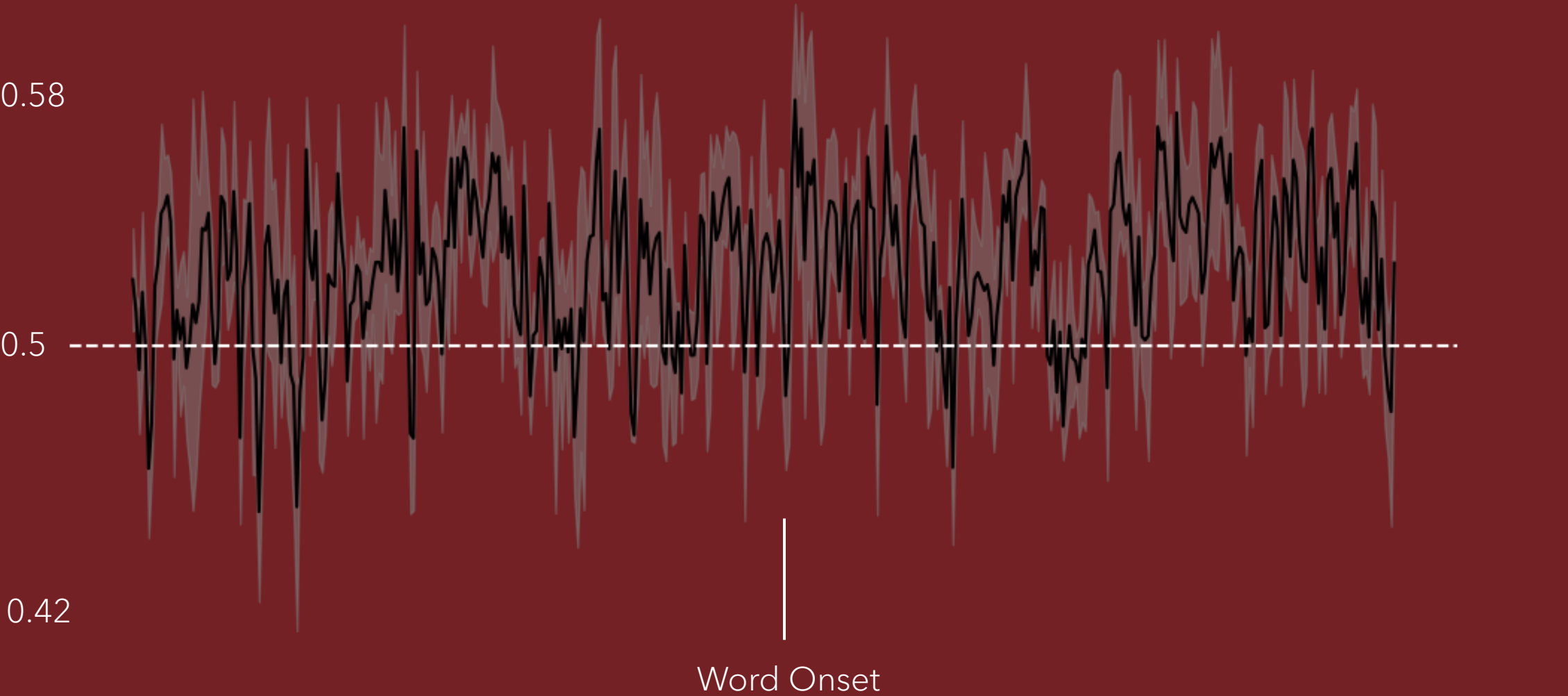
Holmes: Decoding Performance over Time (AUC)



Pronoun (He): Decoding Performance over Time (AUC)



Holmes (He): Decoding Performance over Time (AUC)



ANALYSIS 3: IS THERE A
SEMANTIC TRACE OF THE
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ANALYSIS 3: IS THERE A SEMANTIC TRACE OF THE CHARACTER BEING REPLACED BY THE PRONOUN?



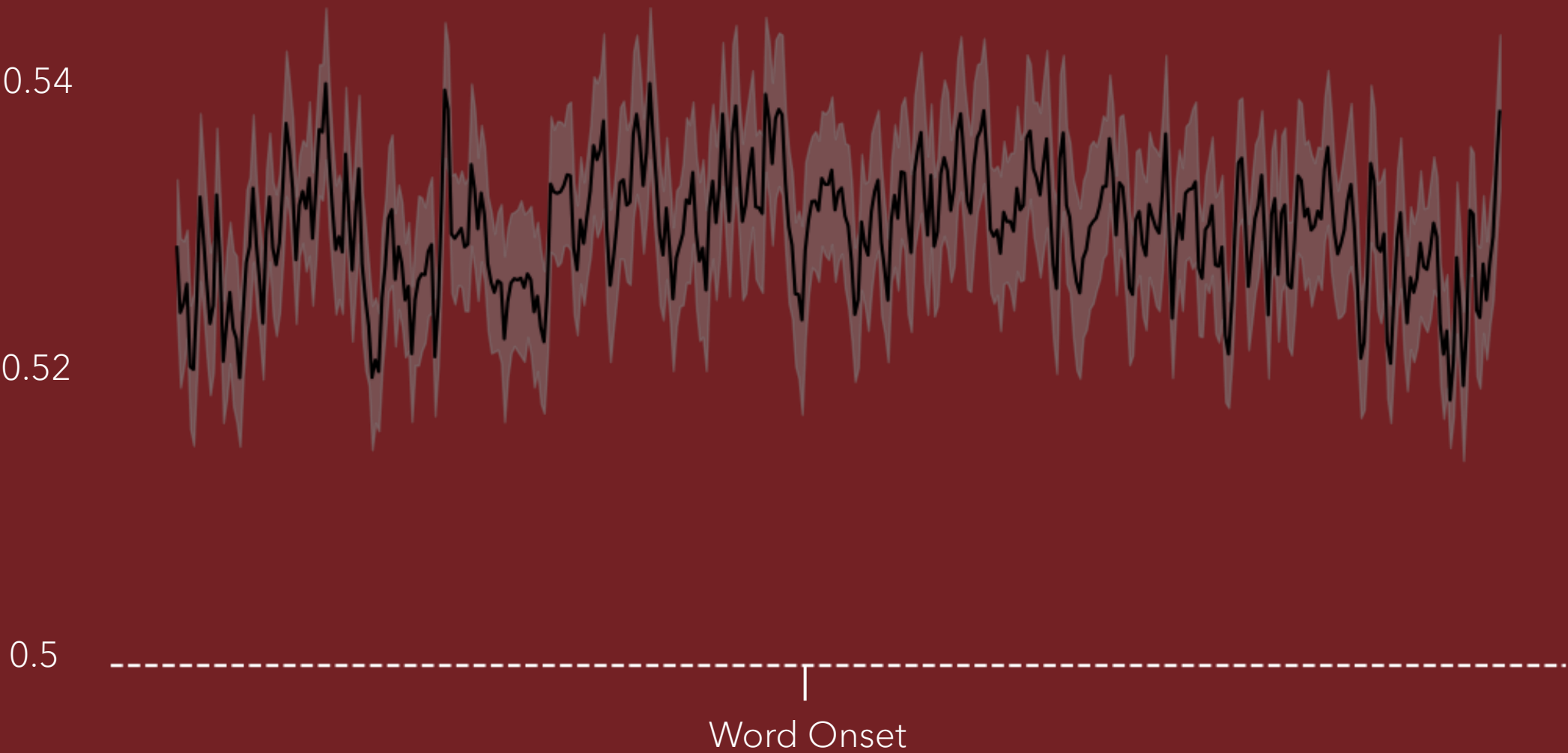
1) Of the previous 18, chose characters with 2 syllables leaving 18

2) Train a LR classifying characters on associated audio stimuli

3) Test on the referential nouns associated with that character

4) Apply same statistical tests

All: Decoding Performance over Time (AUC)

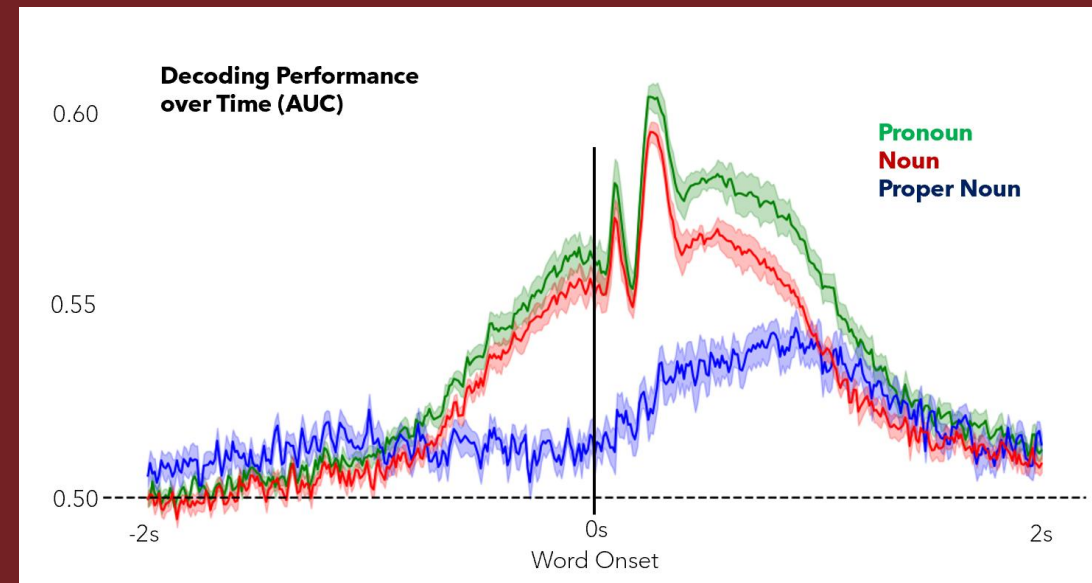


DISCUSSION



ANALYSIS 1

- Clear evidence of different representations between different types of nouns
- Show evidence of prediction for common and pronouns as neural activity becomes decodable pre-onset
- Trend doesn't hold w/ proper nouns



ANALYSIS 2 & 3

- Didn't find evidence of a neural trace of coreference during the processing of pronouns
- Sources of error in assigning coreference – fastcoref model has an F1 score of 78.5 so the data labeling isn't perfect
- Differences in audio stimuli, particularly with experiment 3 add further levels of noise making it challenging to decipher results.
- Despite the dataset being large, certain subtabs of the data are relatively small

Especially characters besides Holmes and Watson

THANK YOU



Any Questions?